



Multi-task Computer Vision on PASCAL VOC2012

Artificial Neural Networks and Deep Learning
Project Report

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Multi-label classification, binary semantic segmentation, and object detection

Abstract

This project develops and evaluates three computer-vision systems on PASCAL VOC2012: multi-label image classification, binary foreground–background semantic segmentation, and 20-class object detection. The experimental protocol separates model development from final assessment. The official training split is divided into training and internal-validation subsets, while the official validation split is reserved as a final holdout and accessed once per task only after the model, checkpoint, threshold, and post-processing rules are fixed.

For classification, three convolutional networks are developed from scratch, followed by three standalone heads on frozen Xception features, an image-level frozen-Xception classifier, and last-block fine-tuning. The frozen-Xception model is selected by internal-validation binary cross-entropy and obtains holdout PR-AUC 0.81444, ROC-AUC 0.95471, and micro-F1 0.75494. For segmentation, a U-Net with skip connections improves internal-validation foreground IoU from 0.46599 for a simple encoder–decoder to 0.51003; its holdout IoU and Dice scores are 0.51007 and 0.67556. For detection, an improved YOLO-style model combines an ImageNet-pretrained Xception backbone, two box slots per 8×8 grid cell, class-aware non-maximum suppression, class balancing, and VOC-compatible handling of difficult objects. It reaches holdout mAP@0.50 of 0.19000.

The results show that transfer learning gives the strongest classification gain, skip connections materially improve foreground segmentation, and object detection remains the most difficult task because of class imbalance, localisation error, and a high false-positive burden. All numbers and figures in this report are traceable to the preserved executed notebook.

Keywords: PASCAL VOC2012; multi-label classification; Xception; transfer learning; U-Net; semantic segmentation; YOLO; object detection.

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1 Introduction

Modern visual recognition systems must answer distinct questions about an image. Classification asks which semantic categories are present, segmentation assigns a class to every pixel, and detection jointly predicts class identity and spatial extent. The project assignment requires all three tasks to be studied using PASCAL VOC2012, with explicit model iteration, design justification, quantitative evaluation, and qualitative visualisation (Bermejo, 2026).

PASCAL VOC provides natural images containing multiple objects, clutter, scale variation, occlusion, and difficult annotations. Its standard categories and evaluation procedures make it a suitable benchmark for comparing classification, segmentation, and detection methods under a common data domain (Everingham et al., 2015). The 20 classes span animals, vehicles, people, and indoor objects, so a single image can contain several positive classification labels and several detection boxes.

The study has four objectives:

1. demonstrate controlled iteration from deliberate overfitting to regularisation in a scratch multi-label classifier;
2. assess three forms of Xception-based transfer learning;
3. compare a simple encoder–decoder with a U-Net for binary semantic segmentation; and
4. develop and validate an improved single-stage YOLO-style detector.

The main methodological contribution is the common evaluation discipline used across tasks. Development decisions are based only on training and internal validation. The official holdout is evaluated exactly once after the winning candidate and all decision rules are locked. This reduces test-set leakage and keeps the final numbers interpretable as out-of-sample performance estimates.

1.1 Architectural foundations

The scratch experiments use dropout, batch normalisation, depthwise-separable convolutions, and residual shortcuts. Dropout reduces co-adaptation (Srivastava et al., 2014); batch normalisation stabilises optimisation (Ioffe and Szegedy, 2015); Xception replaces ordinary convolutions with efficient depthwise-separable operations (Chollet, 2017); and residual shortcuts support gradient propagation in deeper networks (He et al., 2016).

The transfer-learning experiments reuse features learned on ImageNet (Deng et al., 2009). Earlier convolutional features tend to be more general than later task-specific features, motivating a progression from fixed feature extraction to a frozen convolutional base and finally restricted fine-tuning (Yosinski et al., 2014). The segmentation model follows the contracting and expanding paths of U-Net, with same-scale skip connections that recover spatial detail (Ronneberger et al., 2015). The detector follows the single-stage YOLO principle of predicting boxes and class probabilities in one network evaluation (Redmon et al., 2016).

2 Data, Experimental Protocol, and Metrics

2.1 Dataset integrity and official partitions

The attached VOC2012 archive contains 17,125 JPEG images, 17,125 XML annotation files, and 2,913 semantic-segmentation masks. Automated checks found no missing image–XML pairs. The official classification/detection train and validation partitions contain 5,717 and 5,823 images; the official segmentation partitions contain 1,464 and 1,449 images. Official train/validation

overlap is zero for both partition families.

The official training portion of each task is split reproducibly into an 80% training subset and a 20% internal-validation subset. Multi-label-aware iterative allocation balances rare categories across the classification and detection development splits. Segmentation allocation balances semantic-class presence and foreground proportion. The resulting partitions are shown in [table 1](#).

Table 1. Leakage-safe partition sizes used for model development and final assessment.

Task	Training	Internal validation	Final holdout
Multi-label classification	4,574	1,143	5,823
Binary segmentation	1,171	293	1,449
Object detection	4,574	1,143	5,823

For classification, mean labels per image are 1.549, 1.557, and 1.521 in the three partitions. The maximum train-internal-validation class-prevalence gap is 0.629 percentage points, while the maximum shift between the official training pool and holdout is 0.864 percentage points. The close prevalence profiles in [figure 1](#) support the split’s use for model selection.

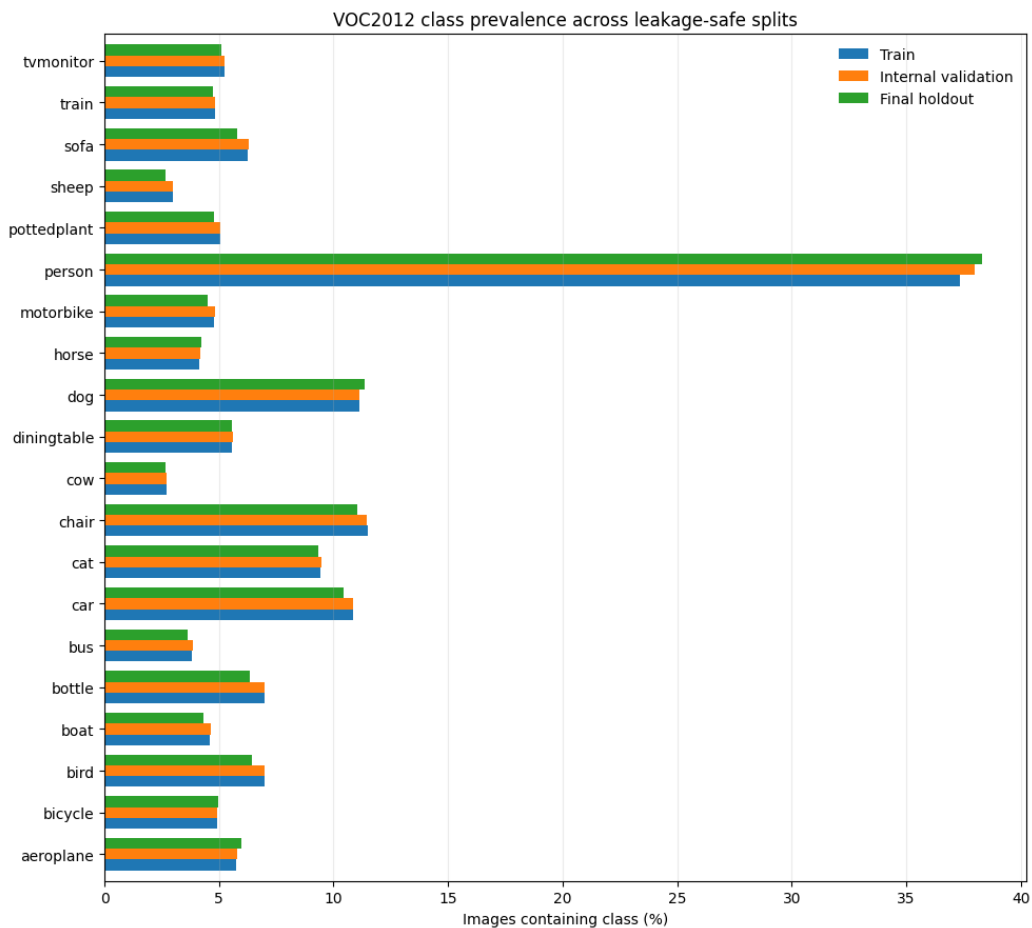


Figure 1. VOC2012 class prevalence across classification training, internal validation, and final holdout. The near-overlapping bars indicate that the development split retains the original multi-label distribution. *Source: preserved executed output from the final project notebook.*

2.2 Preprocessing and reproducibility

All experiments use random seed 42 and deterministic TensorFlow operations. The notebook ran in Kaggle with Python 3.12.13, TensorFlow 2.20.0, Keras 3.13.2, and two NVIDIA Tesla T4 GPUs. Input pipelines use deterministic mapping, batching, prefetching, and shape/range assertions.

Classification images are resized to 180×180 pixels and remain in the raw $[0, 255]$ range in the shared data pipeline. Scratch models rescale once to $[0, 1]$ internally, whereas Xception models apply the required mapping to $[-1, 1]$. Segmentation images and masks are resized to 128×128 pixels. VOC labels 1–20 are merged into foreground, label 0 remains background, and void label 255 receives zero pixel weight. Detection images are resized to 256×256 pixels and encoded on an 8×8 grid with two object slots per cell.

2.3 Evaluation definitions

For N images and $C = 20$ classification labels, predictive binary cross-entropy (BCE) is

$$\text{BCE} = -\frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C [y_{ic} \log(p_{ic}) + (1 - y_{ic}) \log(1 - p_{ic})]. \quad (1)$$

Because positive labels are sparse, PR-AUC is emphasised alongside ROC-AUC; precision–recall analysis is more informative than accuracy or ROC alone under strong imbalance (Saito and Rehmsmeier, 2015). A fixed threshold of 0.50 is used for binary accuracy, precision, recall, F1, Hamming loss, and exact-match accuracy.

For foreground segmentation,

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \quad (2)$$

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} = \frac{2 \text{IoU}}{1 + \text{IoU}}. \quad (3)$$

Dice is an overlap measure originating in set association (Dice, 1945). Foreground IoU is the primary selection metric; Dice, precision, recall, and pixel accuracy provide complementary views. Void pixels do not contribute to the loss or metrics.

For detection, a prediction is a true positive when its class is correct and its intersection-over-union with an unmatched non-difficult ground-truth box is at least 0.50. Predictions best matched to a VOC difficult object are ignored. AP is computed from the interpolated precision–recall curve for each class, and mAP@0.50 is their unweighted mean. Class-aware non-maximum suppression (NMS) uses an IoU threshold of 0.50.

3 Multi-label Image Classification

3.1 Scratch-CNN design sequence

The first experiment is intentionally over-parameterised. Four Conv2D–max-pool blocks with 32, 64, 128, and 256 filters feed a Flatten layer and 512-unit dense head. It has 16,258,900 parameters and omits augmentation, dropout, batch normalisation, and weight decay. The purpose is diagnostic: training should first demonstrate that the model can fit the data, after which generalisation is improved.

The second iteration replaces the large Flatten head with global average pooling and introduces a conventional 32-filter stem followed by four two-layer depthwise-separable blocks with 64, 128, 256, and 512 filters. It combines mild geometric and contrast augmentation, $L_2 = 10^{-4}$ regularisation, batch normalisation, increasing spatial dropout, a 256-unit head, and 0.45 dense dropout. This reduces the model to 681,108 parameters.

The third iteration tests whether residual learning improves the regularised model. Four separable blocks with 64, 128, 256, and 384 filters use 1×1 strided projection shortcuts before element-wise addition. The model has 643,348 parameters and retains augmentation, L_2 regularisation, batch normalisation, global pooling, and dropout. All models use sigmoid outputs, BCE, Adam (Kingma and Ba, 2015), best-checkpoint saving, and early stopping.

The baseline reaches its minimum validation BCE at epoch 3 and then deteriorates to 0.53556 while training loss continues to fall. This is the intended overfitting signal. The regularised model reaches the lowest validation BCE, 0.18454, at epoch 32. The residual model is compact but does not improve the second iteration. The complete comparison is given in table 2.

Table 2. Internal-validation performance of the three scratch-CNN iterations.

Model	Parameters	Best epoch	BCE ↓	Binary accuracy	ROC-AUC	PR-AUC
High-capacity baseline	16,258,900	3	0.22095	0.92511	0.77880	0.26261
Regularised efficient CNN	681,108	32	0.18454	0.93233	0.85991	0.40043
Residual separable CNN	643,348	22	0.20043	0.92878	0.82978	0.33187

Relative to the baseline, the selected regularised model reduces predictive BCE by 0.03641 and increases PR-AUC by 0.13782 while using 95.8% fewer parameters. Residual connections are therefore a meaningful tested hypothesis but a negative result in this setting: additional shortcut structure does not compensate for the different capacity and optimisation dynamics.

3.2 Xception feature extraction and transfer learning

The transfer-learning study follows the three required approaches.

1. **Standalone feature classification.** Frozen Xception with ImageNet weights extracts 2,048-dimensional pooled features. Three heads are compared: a linear sigmoid layer; a 256-unit ReLU layer with 0.45 dropout; and a batch-normalised 256-unit layer with 0.50 dropout followed by a 64-unit bottleneck with 0.25 dropout.
2. **Frozen image-level classifier.** The selected bottleneck architecture is freshly initialised on top of the frozen Xception convolutional base. Training images receive mild augmentation; the base always runs in inference mode.
3. **Last-block fine-tuning.** The frozen classifier checkpoint is the starting point. Only `block14_sepconv1` and `block14_sepconv2` are unfrozen, all Xception BatchNorm layers remain frozen, and the learning rate is reduced to 10^{-5} .

Among standalone heads, the bottleneck obtains the lowest validation BCE (0.10275), although the simpler dropout head has slightly higher PR-AUC (0.79750 versus 0.79445). The predeclared selection criterion is BCE, so the bottleneck architecture is carried forward. The frozen image-level model then improves BCE to 0.10021 and PR-AUC to 0.80487. Fine-tuning changes PR-AUC by only +0.00006 and ROC-AUC by −0.00022, without a BCE advantage. The simpler frozen model is therefore selected.

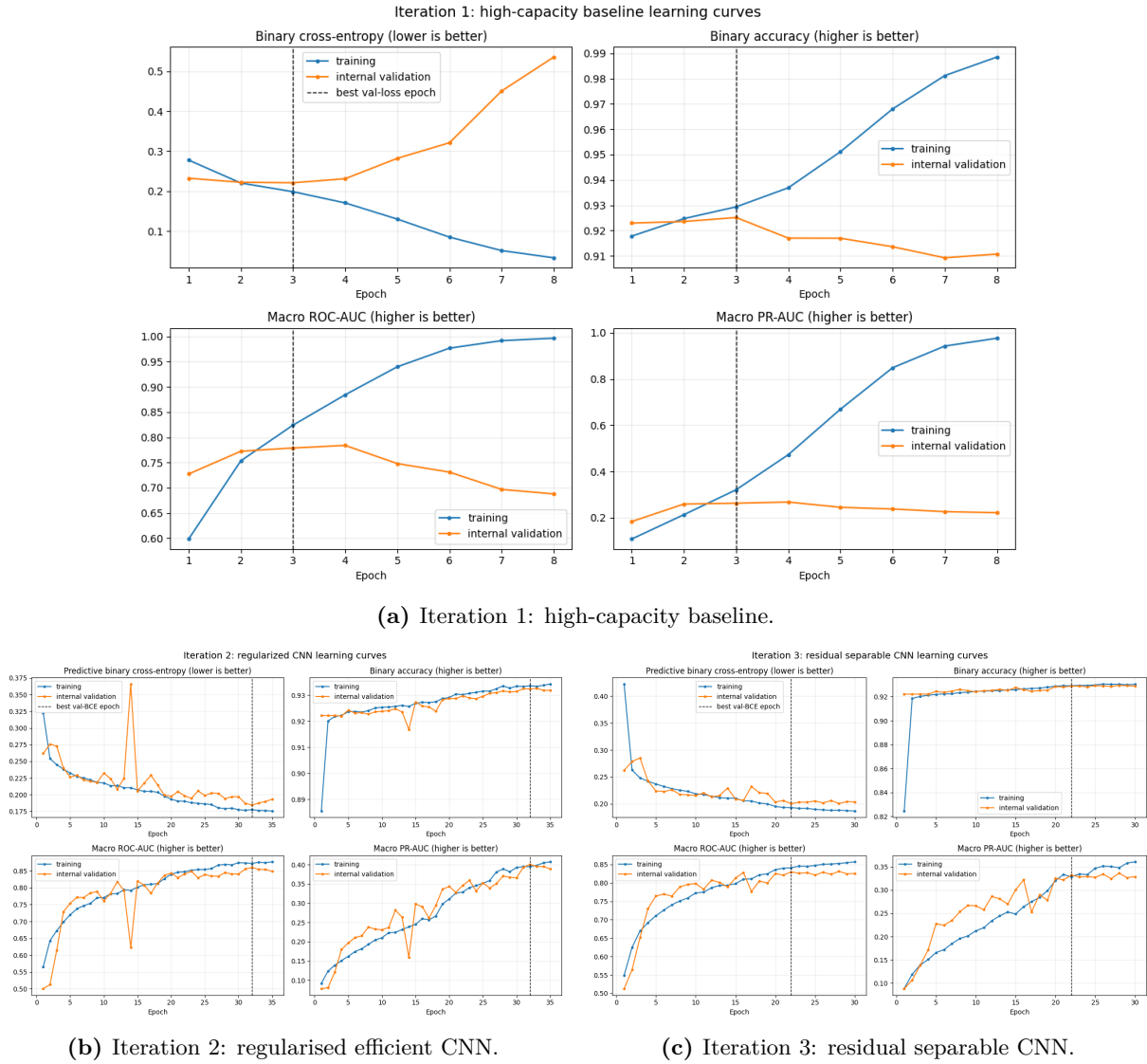
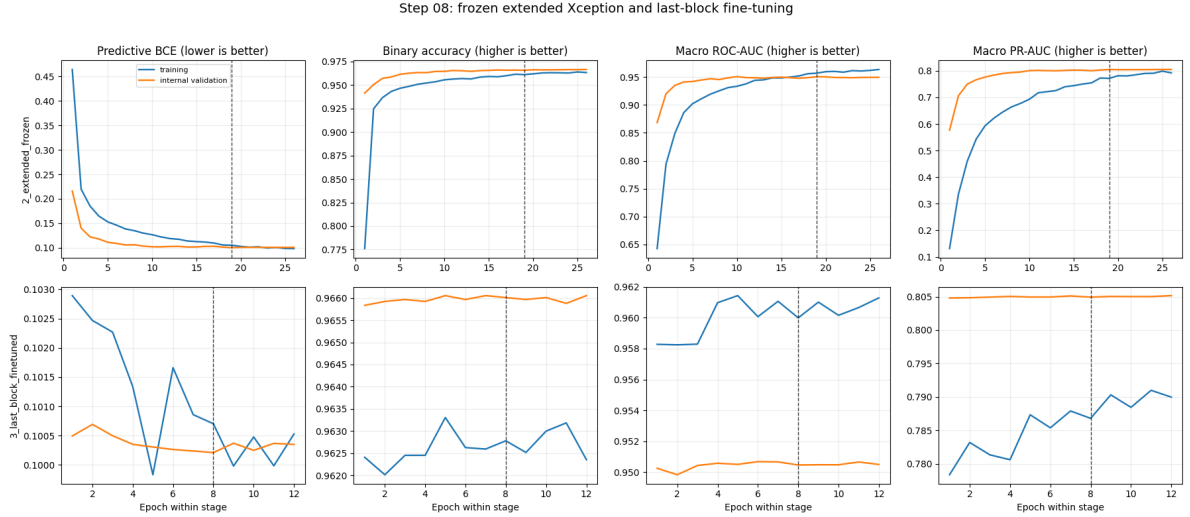


Figure 2. Training and internal-validation learning curves for the three scratch classifiers. The baseline rapidly overfits, while regularisation narrows the train-validation gap. Dashed lines mark the selected epoch in each experiment. *Source: preserved executed output from the final project notebook.*

Table 3. Internal-validation comparison of the required Xception approaches.

Approach	Trainable parameters	Best epoch	BCE ↓	Accuracy	ROC-AUC	PR-AUC
Standalone frozen features	542,548	10	0.10275	0.96492	0.94731	0.79445
Extended frozen Xception	542,548	19	0.10021	0.96570	0.95068	0.80487
Last-block fine-tuned Xception	5,284,180	8	0.10021	0.96601	0.95047	0.80492

**Figure 3.** Frozen-Xception and last-block fine-tuning learning curves. Fine-tuning does not surpass the frozen checkpoint on the selection metric, so the added trainable capacity is not retained. *Source: preserved executed output from the final project notebook.*

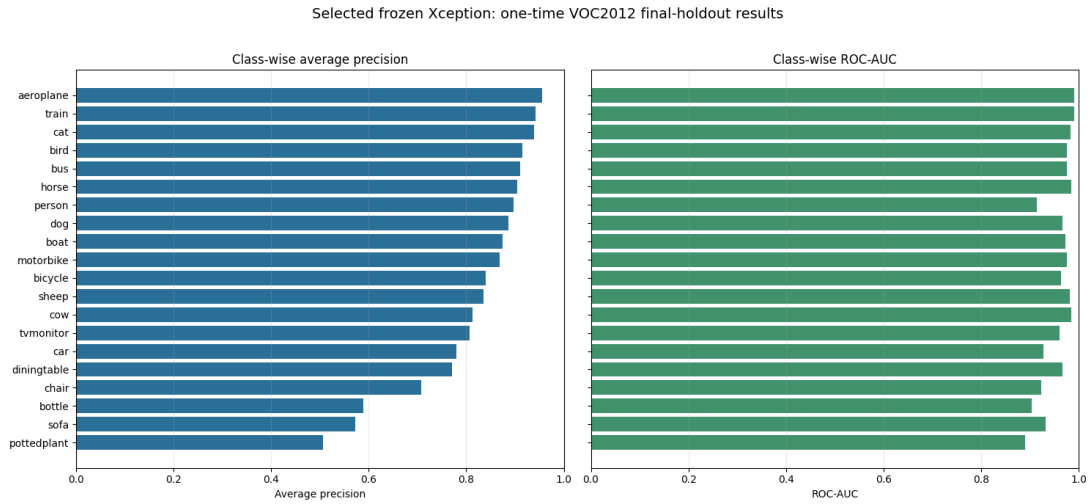
3.3 Locked final classification result

The frozen-Xception classifier is evaluated once on 5,823 holdout images at the fixed threshold 0.50. Its BCE is 0.09539, lower than the validation value by 0.00482. ROC-AUC and PR-AUC improve slightly to 0.95471 and 0.81444, suggesting stable transfer rather than validation-specific selection.

Table 4. Final holdout metrics for the selected frozen-Xception classifier.

Metric	Value	Metric	Value
BCE	0.09539	Macro average precision	0.81558
Binary accuracy	0.96698	Micro average precision	0.84331
ROC-AUC	0.95471	Micro precision at 0.50	0.86634
PR-AUC	0.81444	Micro recall at 0.50	0.66893
Micro F1 at 0.50	0.75494	Macro F1 at 0.50	0.74254
Hamming loss	0.03302	Exact-match accuracy	0.54783

Exact-match accuracy is intentionally stringent because all 20 binary decisions for an image must be correct. The gap between micro precision (0.86634) and recall (0.66893) indicates conservative positive predictions at threshold 0.50. Performance also varies substantially by category. Aeroplane, train, cat, bird, and bus have the strongest AP values (0.95553, 0.94268, 0.93869, 0.91445, and 0.91011), while potted plant, sofa, bottle, chair, and dining table are weakest (0.50612, 0.57237, 0.58912, 0.70707, and 0.77144).



(a) Per-class average precision and ROC-AUC.



(b) Representative holdout predictions at the fixed threshold 0.50.

Figure 4. Final classification evidence. Aggregate performance is strong, but category-level and qualitative results reveal remaining errors that are hidden by binary accuracy. *Source: preserved executed output from the final project notebook.*

4 Binary Semantic Segmentation

4.1 Mask construction and model designs

The segmentation task merges all 20 object categories into a single foreground class. Across the training, validation, and holdout partitions, mean foreground proportions among valid pixels are 26.998%, 27.693%, and 27.215%; mean void proportions are 5.518%, 5.349%, and 5.519%. The matched profiles reduce the risk that model selection is driven by different foreground density.

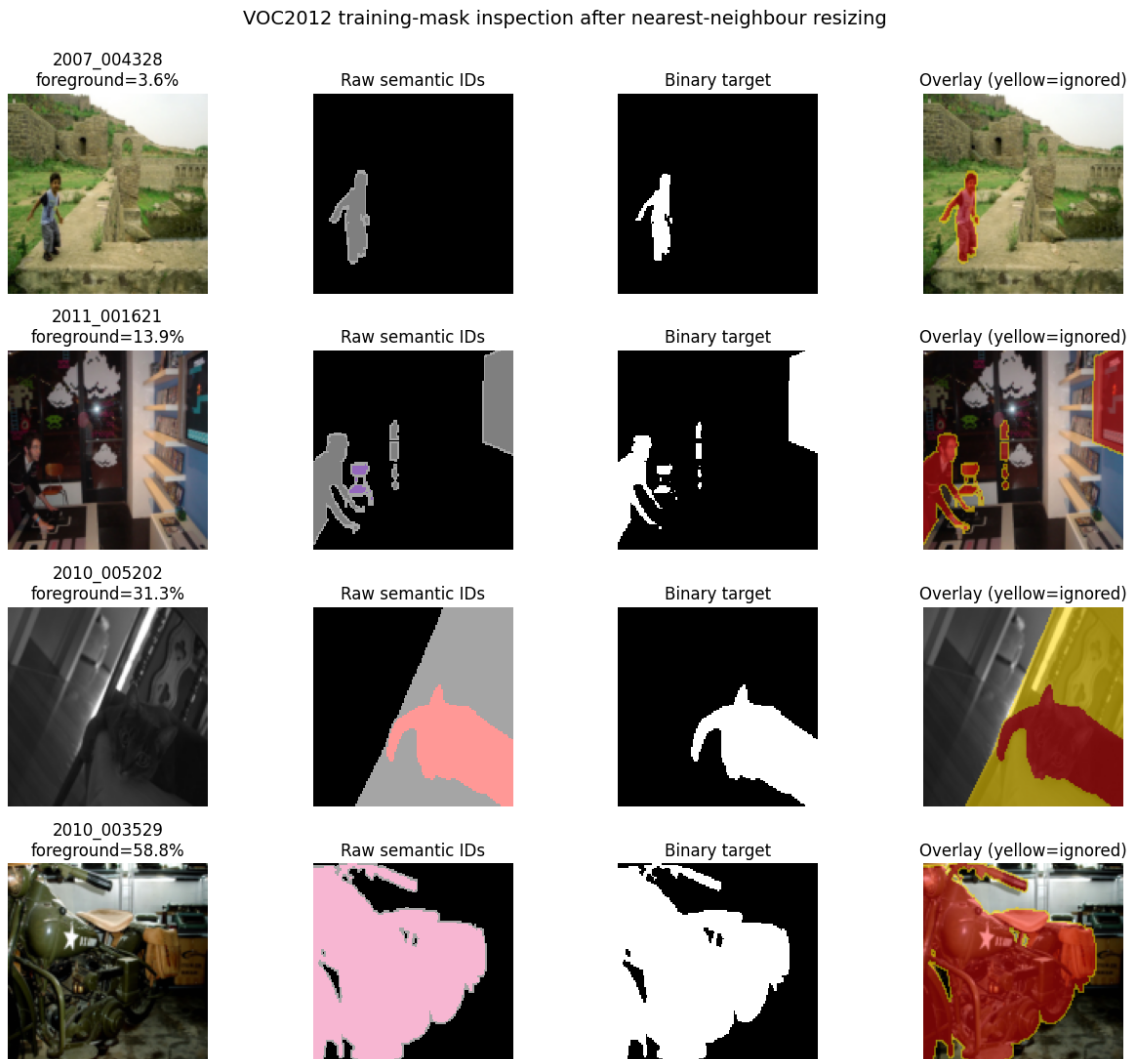


Figure 5. Segmentation preprocessing audit. Each row shows the source image, raw VOC mask, binary foreground target, and void-aware overlay. The visual check confirms alignment and exclusion of label 255. *Source: preserved executed output from the final project notebook.*

The baseline is a three-level encoder–decoder without skip connections. Each level contains two Conv2D layers, the encoder downsamples with max pooling, and the decoder upsamples with transposed convolutions. It has 1,947,105 trainable parameters. Adam uses learning rate 10^{-3} , valid-pixel BCE is the loss, and foreground IoU at threshold 0.50 selects the checkpoint.

The improved model is a three-level U-Net. Conv–BatchNorm–ReLU blocks use 32, 64, 128, and 256 filters. Same-scale encoder features are concatenated into the decoder, and the bottleneck uses 0.30 spatial dropout. A shared horizontal flip transforms image, mask, and void weights together; brightness and contrast jitter affect only images. The model has 1,929,825 total parameters, of

which 1,927,009 are trainable, and uses Adam at 5×10^{-4} .

4.2 Internal-validation comparison

The baseline reaches foreground IoU 0.46599 at epoch 15. U-Net reaches 0.51003 at epoch 15, an absolute improvement of 0.04404, while also improving precision from 0.57314 to 0.62670 and recall from 0.71368 to 0.73260. The gain with slightly fewer parameters supports the architectural explanation: skip connections restore fine-scale boundary information lost during downsampling.

Table 5. Internal-validation comparison of the segmentation candidates.

Model	Total parameters	Best epoch	Accuracy	Foreground IoU	Precision	Recall
Encoder-decoder baseline	1,947,105	15	0.77695	0.46599	0.57314	0.71368
U-Net with skip connections	1,929,825	15	0.80806	0.51003	0.62670	0.73260

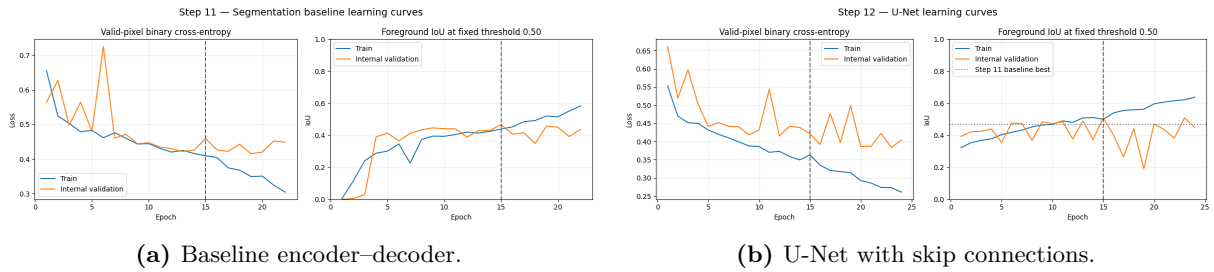


Figure 6. Segmentation learning curves. U-Net achieves the higher locked validation IoU despite substantial epoch-to-epoch variability, motivating checkpoint selection by IoU rather than final-epoch loss. *Source: preserved executed output from the final project notebook.*

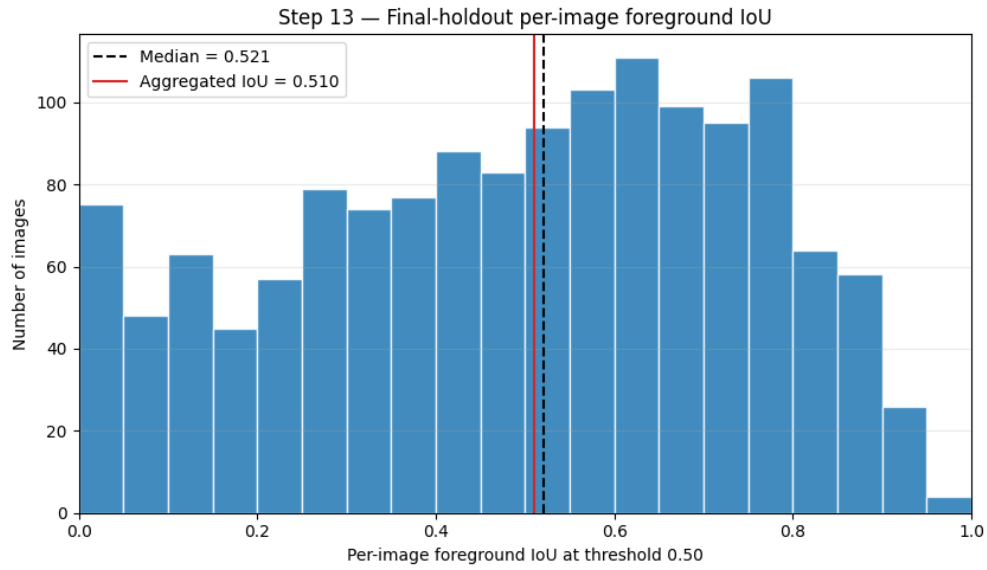
4.3 Locked final segmentation result

The selected U-Net is evaluated once on 1,449 holdout images. Foreground IoU is 0.51007, only 0.00004 above internal validation. Dice/F1 is 0.67556, precision is 0.62766, and recall is 0.73138. Pixel accuracy is 0.81227, but it is secondary because abundant background pixels make it easier to obtain a high value without accurate object boundaries.

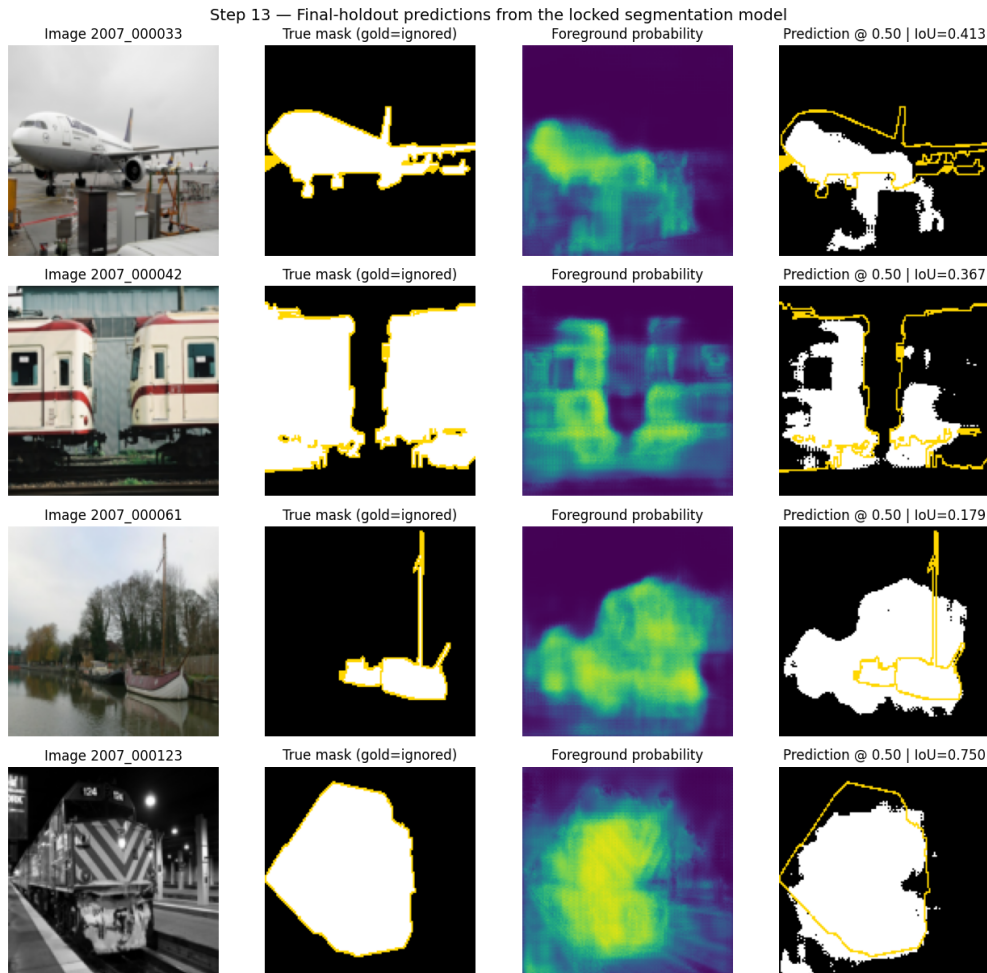
Table 6. Final holdout metrics for the selected U-Net.

Metric	Value	Metric	Value
Valid-pixel BCE loss	0.40969	Foreground IoU	0.51007
Binary pixel accuracy	0.81227	Dice/F1	0.67556
Foreground precision	0.62766	Foreground recall	0.73138
Mean per-image IoU	0.49162	Median per-image IoU	0.52053
First quartile IoU	0.29835	Third quartile IoU	0.69514

The aggregate IoU is stable, but the per-image interquartile range (0.29835–0.69514) is wide. Qualitative examples show that the model often captures the main object extent but smooths thin structures and can merge nearby foreground regions. These errors are consistent with resizing masks to 128×128 and reducing all semantic classes to one foreground channel.



(a) Distribution of per-image foreground IoU.



(b) Representative holdout masks and IoU scores.

Figure 7. Final segmentation evidence. The distribution exposes heterogeneous image difficulty, while the examples compare image, target, predicted probability, thresholded mask, and boundary error. *Source: preserved executed output from the final project notebook.*

5 Object Detection

5.1 Target encoding and detector design

Detection uses the 11,540 official classification/detection train-validation images. Each easy object is assigned to the grid cell containing its centre. A slot stores objectness, within-cell centre offsets, square-root relative width and height, and a 20-class one-hot vector. Two slots allow two easy-object centres in the same cell; difficult-only cells receive an ignore mask for objectness. Across the development set, 13,528 of 13,609 easy boxes are encoded, giving 99.405% coverage. Collisions drop 62 training and 19 internal-validation boxes; 78 boxes are unrepresentable in the holdout encoding.



Figure 8. Audited two-slot YOLO target for a crowded training image. Red boxes show easy objects, dashed grey boxes show difficult objects, and the overlaid grid makes target assignment and collisions inspectable. *Source: preserved executed output from the final project notebook.*

The improved detector uses ImageNet-pretrained Xception at 8×8 spatial resolution. A 512-filter and 256-filter separable-convolution head with batch normalisation, ReLU, $L_2 = 10^{-5}$, and 0.25 dropout predicts $8 \times 8 \times 2 \times 25$ raw outputs. Photometric augmentation uses contrast jitter and Gaussian noise; geometric augmentation is deliberately omitted because it would require equivalent box-target transformations. The network has 22,080,090 parameters. During the six-epoch warm-up, 1,217,074 parameters are trainable. During restricted block-14 fine-tuning, 5,958,706 parameters are trainable and all pretrained BatchNorm statistics remain frozen.

The custom loss is

$$\mathcal{L} = \mathcal{L}_{\text{obj}} + 5\mathcal{L}_{\text{box}} + \mathcal{L}_{\text{class}}. \quad (4)$$

Negative objectness slots receive weight 0.10 to control their numerical dominance. The class loss uses inverse-frequency weights clipped to $[0.5, 2.5]$. Box loss is squared error over centre offsets and square-root dimensions, evaluated only on positive slots. At inference, objectness and

class probabilities are multiplied, boxes are decoded to normalised coordinates, and class-aware NMS is applied.

5.2 Internal-validation model selection

The detector is selected directly by mAP@0.50 rather than the multi-component training loss. During frozen-backbone warm-up, validation mAP rises from 0.01692 at epoch 1 to 0.21395 at epoch 4, then declines. Restricted fine-tuning reaches at most 0.20900 and therefore does not beat the warm-up checkpoint. Restoring epoch 4 gives slot-objectness precision 0.17021, recall 0.59196, and positive-slot class accuracy 0.72285.

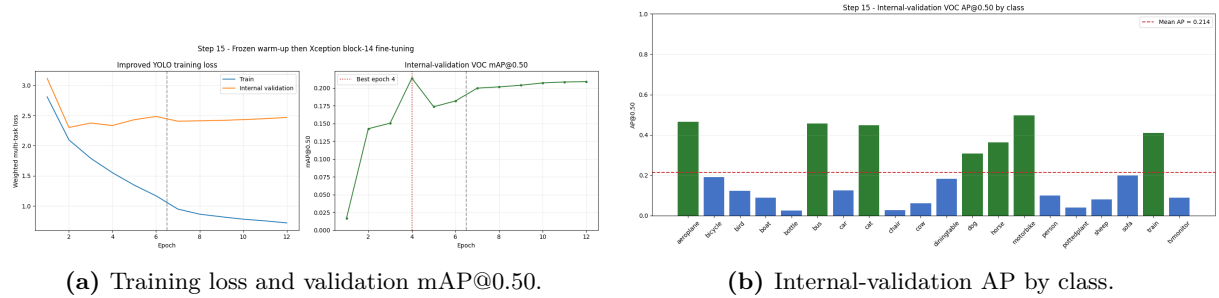


Figure 9. Detector model-selection evidence. The best checkpoint occurs before fine-tuning; class-level AP reveals large heterogeneity around mean mAP 0.21395. *Source: preserved executed output from the final project notebook.*

5.3 Locked final detection result

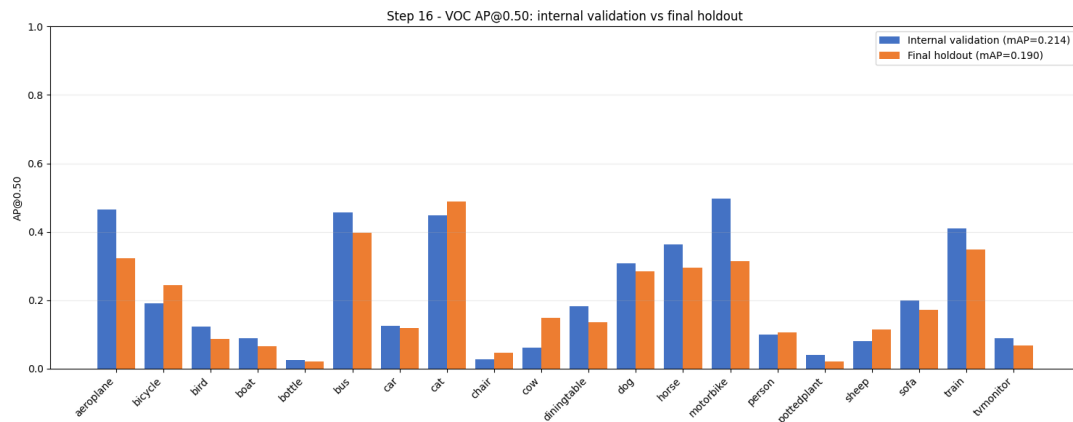
One inference pass over the 5,823-image holdout produces mAP@0.50 of 0.19000, a decrease of 0.02395 from internal validation. The weighted multi-task loss is 2.30964, with objectness, box, and class components 0.24862, 0.22318, and 0.92023. Slot-objectness recall is 0.59776, substantially higher than precision 0.16975, and positive-slot class accuracy is 0.72942. These slot metrics diagnose the raw grid output; final class-specific AP remains the primary detector measure.

Table 7. Final holdout metrics for the improved YOLO-style detector.

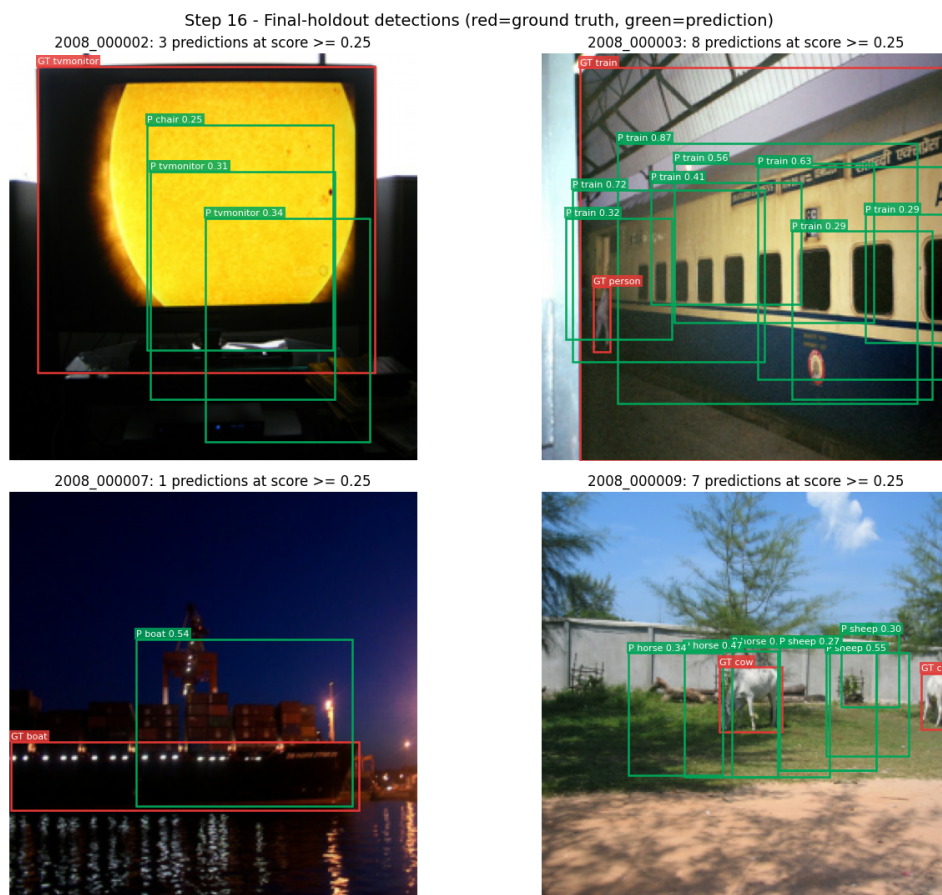
Metric	Value	Metric	Value
Weighted multi-task loss	2.30964	mAP@0.50	0.19000
Objectness loss	0.24862	Slot precision at 0.50	0.16975
Box loss	0.22318	Slot recall at 0.50	0.59776
Class loss	0.92023	Positive-slot class accuracy	0.72942

Cat, bus, train, aeroplane, and motorbike are the five strongest holdout classes, with AP 0.48907, 0.39775, 0.34725, 0.32349, and 0.31497. Potted plant, bottle, chair, boat, and television monitor are the weakest, with AP 0.02012, 0.02149, 0.04623, 0.06618, and 0.06851. Small objects and crowded indoor categories are especially difficult for the coarse grid and fixed input resolution.

6 Cross-task Discussion and Validity



(a) Per-class AP@0.50 on internal validation and final holdout.



(b) Representative final-holdout detections at visual score threshold 0.25.

Figure 10. Final detection evidence. Per-class comparisons show which changes generalise, while qualitative examples expose duplicate or misplaced proposals that aggregate mAP alone cannot explain. *Source: preserved executed output from the final project notebook.*

6.1 What the comparisons establish

The strongest result is the classification transfer-learning gain. Relative to the selected scratch model, frozen Xception reduces validation BCE from 0.18454 to 0.10021 and doubles PR-AUC from 0.40043 to 0.80487. This supports the practical value of transferable ImageNet features for a moderately sized natural-image dataset. Fine-tuning does not improve the locked selection metric, showing that greater trainable capacity is not automatically beneficial.

For segmentation, U-Net’s 0.04404 validation-IoU improvement over the baseline is achieved without more total parameters. The evidence therefore isolates skip connections and the associated training regularisation as more plausible drivers than raw capacity. The almost identical validation and holdout IoU indicates stable aggregate generalisation, although the per-image distribution remains broad.

Detection is methodologically complete but quantitatively weaker. The detector performs classification, objectness estimation, and localisation jointly under a coarse spatial encoding. High objectness recall with low precision implies many false-positive proposals; class AP then suffers further from localisation and classification errors. The negative fine-tuning result also suggests that the main bottleneck lies in the head, target encoding, data scale, and post-processing, not merely in the last Xception block.

6.2 Robustness and leakage controls

The following checks strengthen the credibility of the results:

- official holdouts are never used for epoch selection, architecture comparison, threshold tuning, or NMS tuning;
- each task logs zero overlap between development and holdout identifiers;
- split-prevalence gaps are quantified before modelling;
- preprocessing ranges and output shapes are asserted before training;
- best checkpoints are restored and evaluated independently;
- frozen and fine-tuned Xception weight boundaries are fingerprint-audited;
- void segmentation pixels and difficult detection objects are explicitly excluded according to their intended semantics; and
- every headline result is supported by preserved executed output rather than a reconstructed calculation.

The small validation–holdout changes for classification and segmentation support the split design. Detection has a larger, but still plausible, mAP decrease. The direction and class-specific variation are disclosed rather than hidden.

6.3 Limitations

Several limitations constrain interpretation.

1. **Single split and seed.** Deterministic execution improves reproducibility but does not quantify variability across seeds or alternative development splits.
2. **Fixed thresholds.** A universal 0.50 classification and segmentation threshold simplifies fair holdout evaluation but is not necessarily optimal for every class or deployment cost.
3. **Resolution.** Resizing to 180×180 , 128×128 , and 256×256 limits small-object and fine-boundary fidelity.
4. **Binary segmentation.** Merging 20 categories removes class-specific segmentation information and makes touching objects a single foreground region.

5. **Detection target capacity.** The two-slot 8×8 target represents 99.405% of development easy boxes, but collisions still discard 81 development boxes and 78 holdout boxes.
6. **Detector calibration.** Low proposal precision shows that objectness and post-processing are not well calibrated. The visual threshold is used only for display; AP uses the locked low score floor and full ranking.
7. **Compute budget.** Architecture search is deliberately bounded to the assignment requirements and available dual-T4 time. No large ensemble or extensive hyperparameter sweep is performed.

7 Conclusion and Future Work

This project satisfies the three requested computer-vision tasks under one leakage-aware experimental protocol. Controlled scratch-CNN iteration demonstrates the transition from overfitting to regularisation. Frozen-Xception transfer learning produces the best classifier, with holdout PR-AUC 0.81444 and ROC-AUC 0.95471; restricted fine-tuning adds cost without measurable validation benefit. For segmentation, U-Net improves on a parameter-matched encoder–decoder and achieves holdout IoU 0.51007 and Dice 0.67556. The improved YOLO-style detector reaches holdout mAP@0.50 of 0.19000 and reveals detection as the principal remaining challenge.

Future improvements should follow the observed failure modes. Classification could use class-specific thresholds selected only on internal validation and report their calibration benefit. Segmentation could preserve higher spatial resolution and use losses that directly emphasise overlap or boundaries. Detection would benefit most from multi-scale features, anchor or anchor-free heads with greater spatial capacity, joint geometric box augmentation, focal-style objectness treatment, and stricter internal-validation calibration of proposal scores. These extensions should retain the same locked-holdout protocol so that improvements remain credible.

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A Assignment Requirement Coverage

Table 8. Traceability from explicit assignment requirements to report evidence.

Assignment requirement	Evidence in this report
Scratch CNN defined layer-by-layer	Three original architectures and their design rationale in table 2 and figure 2 .
At least three scratch iterations	High-capacity baseline, regularised efficient CNN, and residual separable CNN.
First overfit, then regularise	Deliberate overfitting diagnosis and regularisation comparison in figure 2 .
Advanced classification features	Callbacks, augmentation, dropout, batch normalisation, depthwise-separable convolutions, and residual connections are all implemented and discussed.
Xception standalone feature extraction	Three dense-head iterations and selected bottleneck architecture in table 3 .
Frozen Xception image classifier	Extended frozen-base classifier, validation comparison, and final holdout result.
Fine-tune last Xception block	Block-14 fine-tuning from the frozen checkpoint with BatchNorm frozen; negative result discussed.
Binary foreground-background segmentation	Void-aware binary mask construction and split audit in figure 5 .
Build and compare segmentation output with ground truth	Baseline plus U-Net, learning curves, final masks, and boundary errors in figure 7 .
Reflect on Dice and other segmentation metrics	IoU, Dice, precision, recall, pixel accuracy, and per-image IoU distribution are defined and interpreted.
Develop and validate an improved detector	Xception-backed two-slot YOLO-style model, custom loss, class-aware NMS, difficult-object handling, mAP selection, per-class AP, and final examples.
Discuss design choices and results	Each task contains methods, quantitative comparison, interpretation, negative findings, and limitations.
Include relevant numbers and figures	Ten evidence-led figure groups and eight quantitative tables are included; all come from executed notebook outputs.

B Reproducibility Record

Table 9. Software, hardware, and locked evaluation settings.

Item	Recorded setting
Platform	Kaggle notebook
Python / TensorFlow / Keras	3.12.13 / 2.20.0 / 3.13.2
Accelerator	Two NVIDIA Tesla T4 GPUs
Random seed	42
Determinism	TensorFlow deterministic operations enabled
Classification input / batch	180×180 / 32
Segmentation input / batch	128×128 / 16
Detection input / batch	256×256 / 16
Classification threshold	0.50, fixed before holdout
Segmentation threshold	0.50, fixed before holdout
Detection AP and NMS IoU	0.50 / 0.50
Detector AP score floor	0.001
Detector visual score threshold	0.25; visualisation only
Holdout use	One selected model and one inference pass per task; no post-holdout tuning

The submitted notebook contains the complete preprocessing, model definitions, training histories, checkpoint-selection logic, evaluation code, metric tables, and figures. Computational Steps 1–16 were preserved during final editorial cleanup and were not rerun or altered.

C Acknowledgement of Generative-AI Assistance

OpenAI was used as a supporting tool for requirements checking, report structuring, language editing, LaTeX formatting. The reported experiments, numerical values, tables, and figures are derived from the executed notebook outputs. The author reviewed the final content and remains responsible for the methodological choices, interpretation, and submitted work.